NELSON KOSKELA

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience building and operating reliable backend and data infrastructure in Python and SQL within a high-volume, compliance-driven financial environment. Track record of treating correctness as a first-class requirement: designed and led a parallel 'prod vs. prod' validation framework with automated reconciliation to catch inconsistencies between platforms before they reached production. Comfortable partnering directly with stakeholders to turn ambiguous requirements into dependable, well-tested systems, with hands-on experience applying agentic engineering to automate operational workflows.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C#, Swift, HTML/CSS

Data & Analytics: Amazon Athena, PostgreSQL, SQLite3 - schema design & query optimization

Cloud & Infrastructure: AWS, Docker, Unix/Linux, Git

Frameworks & Libraries: Flask, React, Bootstrap, Ruby on Rails

Methodologies: Agile/Scrum, Object-Oriented Programming, Distributed Systems, Data Reconciliation

AI & Agentic Tooling: Agentic engineering, AI-agent orchestration, AI-assisted development workflows

PROFESSIONAL EXPERIENCE

J.P. Morgan Chase & Co. - Software Engineer II, Corporate & Investment Banking — *July 2022 - Present*

Houston, TX

- Engineered and operated data pipelines and analytics infrastructure supporting regulatory and risk reporting, reliably processing multi-terabyte datasets across global business units in a production, always-on environment.
- Partnered directly with business stakeholders to build new reporting and infrastructure for enterprise risk-data systems, delivering data in live, end-of-day (EOD), and start-of-day (SOD) formats.
- Designed and led a parallel 'prod vs. prod' validation framework for safely migrating infrastructure between platforms, with automated reconciliation ensuring the two environments matched to the record before cutover.
- Applied agentic engineering - orchestrating AI-driven workflows to analyze large volumes of data and conduct deep dives into the Athena ecosystem - cutting support workload by 50%.
- Partnered with downstream and cross-functional Agile teams to enforce data quality, security, and regulatory compliance, operating within strict access-control, auditability, and governance requirements.

StreetCode Academy - Diversity & Inclusion Officer & Freelance Developer — *Fall 2020 - Fall 2022*

Remote

- Developed responsive web applications using HTML, CSS, and JavaScript, enhancing customer engagement and usability.
- Applied Agile Scrum methodology to plan sprints, track progress, and deliver client solutions on time.
- Managed logistics and database tracking systems for internal inventory, improving item tracking accuracy by 50% by building and delivering full, accurate analytics reporting.

PROJECTS

Personal Portfolio Site - *nelsonkoskela.dev*

- Applied agentic engineering to design and build "Pipeline World," a live SDLC visualizer that streams pipeline runs in real time over WebSockets - authoring detailed specs and orchestrating a coding agent through implementation, end to end.
- Independently scoped, built, and shipped a containerized (Docker) Flask/Python portfolio, including an SRE/Infra layer demonstrating Redis-based queueing and caching.

Beeznest - *github.com/nellykelly/BezzNest*

- Built a B2B networking platform using Ruby on Rails and SQLite3, architecting the data layer to facilitate inter-company communication.
- Awarded 2nd place at StreetCode Accelerator Demo Day for product innovation and design; continued development into market research and business strategy.

EDUCATION

Cogswell Polytechnical Institute - B.S. in Computer Science — 2019 - 2022

Relevant Coursework: Data Structures, Database Systems, Linux Programming, Software Engineering Methods

Howard University - B.S. in Computer Science (Transferred) — 2017 - 2019

Relevant Coursework: Large Scale Programming, Game Design, Affective Computing